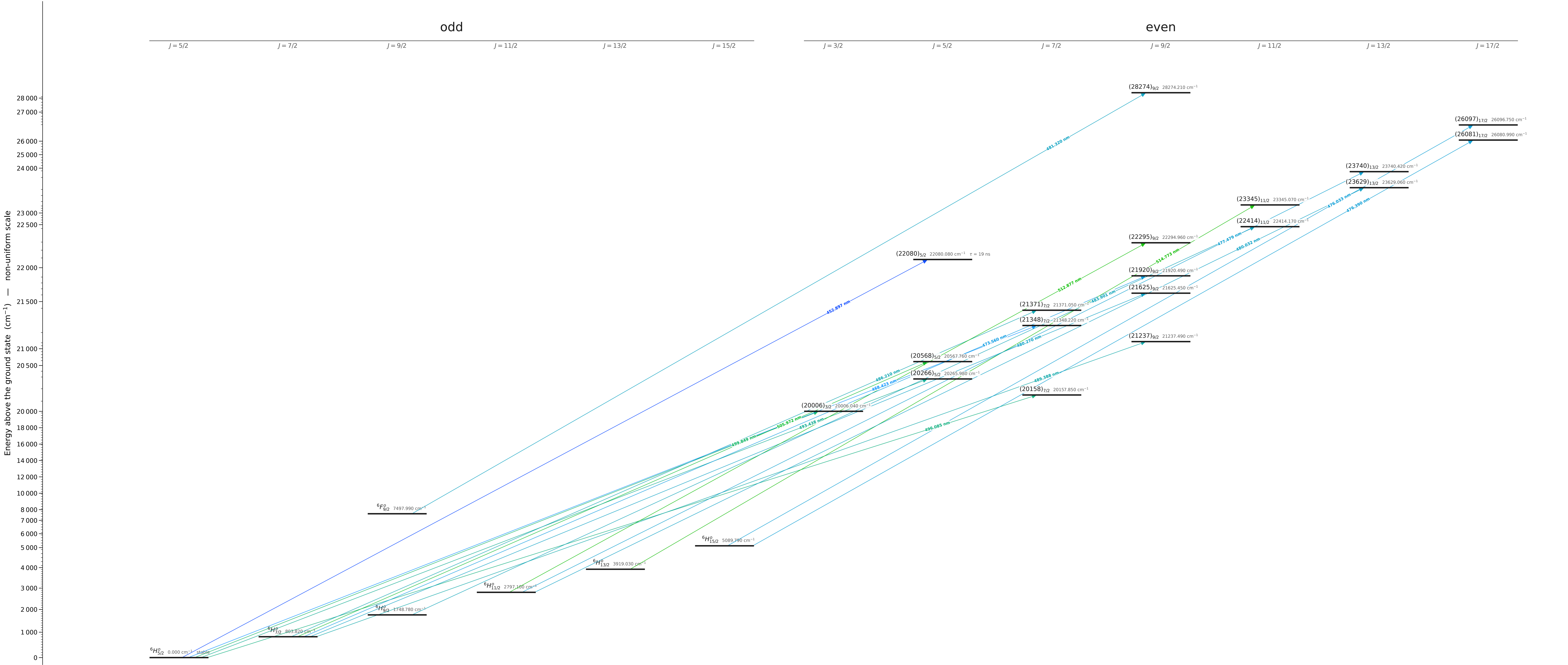
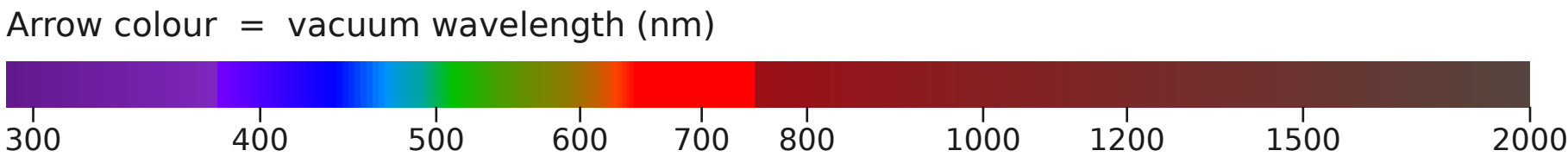


Promethium

Pm (Pm I). Strongest classified transitions below 2 μm (NIST): 25 levels, 18 lines.

Pm ⁺ ionisation limit 45020.80 cm⁻¹ = 5.58187 eV (222.120 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element |(U||er||J')| in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	(J er J') (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
⁶ H _{5/2} ^o - (22080) _{5/2}	452.8969	452.7699	661.9442	-	-	8.377	-	first step of the Mainz three-step resonance-ionisation s...
⁶ H _{5/2} ^o - (21348) _{7/2}	468.4231	468.2921	640.0035	-	-	-	-	first step of the Mainz three-step resonance-ionisation s...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

¹⁴⁷Pm hyperfine structure (I = 7/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)						
⁶ H _{5/2} ^o	620.3(14)	-407(18)	F=1: -7196.41	F=2: -5839.52	F=3: -3847.80	F=4: -1273.58	F=5: +1813.39	F=6: +5325.88	
⁶ H _{7/2} ^o	447(9)	268(71)	F=0: -6898.52	F=1: -6478.72	F=2: -5633.65	F=3: -4352.40	F=4: -2618.59	F=5: -410.39	F=6: +2299.51 F=7: +5543.85
(21348) _{7/2}	438.2(15)	-48(13)	F=0: -6927.36	F=1: -6484.27	F=2: -5599.05	F=3: -4273.67	F=4: -2511.08	F=5: -315.18	F=6: +2309.12 F=7: +5355.95
(22080) _{5/2}	500.0(14)	-119(15)	F=1: -5688.75	F=2: -4654.75	F=3: -3116.50	F=4: -1089.30	F=5: +1406.45	F=6: +4345.25	

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
⁶ H _{5/2} ^o - (22080) _{5/2}	452.897	147-146	226(12)
⁶ H _{5/2} ^o - (22080) _{5/2}	452.897	147-145	344(10)
⁶ H _{5/2} ^o - (22080) _{5/2}	452.897	147-144	594(10)
⁶ H _{5/2} ^o - (22080) _{5/2}	452.897	147-143	737(11)
⁶ H _{5/2} ^o - (21348) _{7/2}	468.423	147-146	-164(16)
⁶ H _{5/2} ^o - (21348) _{7/2}	468.423	147-145	-216(11)
⁶ H _{5/2} ^o - (21348) _{7/2}	468.423	147-144	-384(13)
⁶ H _{5/2} ^o - (21348) _{7/2}	468.423	147-143	-495(14)

Shift = ν(first isotope) - ν(second isotope).